

Università di Pisa

Dipartimento di Ingegneria dell'Informazione

Master's Degree in Artificial Intelligence and Data Engineering

Master's Thesis

Data Analysis and Data Mining on Decentralised Prediction Markets: the Price Process of Polymarket from One Minute to Twelve Hours

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Abstract

This thesis studies Polymarket, a decentralised prediction market, as a data-analysis and data-mining object: how twenty-one days of order-book, on-chain transaction and market data were collected and prepared, what the price process and the transaction graph look like, and which families of models can predict the direction of a market's price over horizons from one minute to twelve hours.

[TODO: rewrite at the end, together with the introduction]

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Chapter 1

Introduction

[TODO: to be written last, once the other chapters are stable. Only the outline for now.]

1.1 Context

Prediction markets as aggregators of dispersed information [1, 2]; a decentralised one, Polymarket, exposes the whole order book and every trade with the wallet behind it, at a resolution of about two seconds, on tens of thousands of markets at once. [TODO: one page on why this is a different object for a data scientist; the mechanics stay in Section 2.2]

1.2 The problem

[TODO: the task (direction of the mid price over one minute to twelve hours) and why it is hard: a price consolidates information; it is bounded and discrete; it is mostly still; much of what moves it is outside the data]

1.3 Research questions

Q1 What regularities does the price process of a prediction market have, and how is its trading organised?
Q2 Which publicly observable information predicts the direction of the mid price?
Q3 Which families of models can read that information, and what is a sequential state or a neighbourhood worth over a stateless learner on the same inputs?
Q4 How do the answers change with the horizon and with the kind of market?

1.4 Contributions

[TODO: list, once the results chapter is in place]

1.5 Structure of the thesis

[TODO: one paragraph per chapter]

Chapter 2

Background

This chapter gives the background that the rest of the thesis assumes: what a prediction market is and how Polymarket works (Sections 2.1 and 2.2), and the notions of machine learning that the methodology uses: time series and sequence models (Section 2.3), graphs and graph neural networks (Section 2.4), learners on tables of features (Section 2.5) and evaluation under class imbalance (Section 2.6).

2.1 Prediction markets

A prediction market trades contracts whose value is tied to the outcome of a future event. The simplest contract is the *binary* one: it pays one unit if the event happens and nothing otherwise. If traders are risk-neutral and have no budget constraints, the price of such a contract equals the market's aggregate probability of the event; under weaker assumptions the price is still close to the mean belief of the traders [1, 3]. Surveys of the empirical literature find that market prices are well calibrated and competitive with polls and expert forecasts, and that they react to news within minutes [1, 2, 4].

[TODO: the favourite–longshot bias [5] and its Polymarket version; the concentration of profits in a few accounts]

2.2 Polymarket

2.2.1 How Polymarket works

Polymarket is a decentralised prediction market built on the Polygon blockchain, with the stablecoin USDC as collateral. Every market is a question with two outcomes: "Yes" and "No" for a binary question, "Up" and "Down" for a price interval, the two teams of a game, "Over" and "Under" for a total. Each outcome is a conditional token: one dollar of collateral can at any time be *split* into one Yes token and one No token, a pair can be *merged* back into one dollar, and after resolution the winning token is *redeemed* for one dollar and the losing one for nothing. The two prices therefore sum to about one dollar, and a trader who wants to sell Yes at price p is indistinguishable, for the matching engine, from one who wants to buy No at $1 - p$.

Trading takes place on a central limit order book that is *hybrid*. Orders are signed messages that the trader submits to an off-chain operator, which keeps the book, publishes every change to it and matches crossing orders; a match is then settled on-chain by the exchange contract, which transfers tokens and collateral and emits a fill event with the maker, the taker, the token, the side, the quantity, the price and the fee. The book is thus

observable as a stream of price-level updates, and the trades as on-chain records attributed to wallet addresses.

[TODO: multi-outcome events (the candidates of an election, the strikes of a ladder) and the bounds between the prices of sibling contracts [6]]

2.2.2 Tokens, contracts and transactions: the blockchain layer

Polygon produces a block roughly every two seconds; every fill is a transaction in one of its blocks and carries the block number, the transaction hash and the wallet addresses of the maker and the taker. Because settlement is on-chain, every trade is attributed to a wallet that persists across markets and across time, which makes it possible to build a bipartite graph between wallets and markets. [TODO: what wallet-level data has been used for on Polymarket; the block as the time step of the models]

2.2.3 The limit order book

A limit order book [7] is, for one token, two sorted lists of resting limit orders, bids and asks, each with the total size resting at that price. The highest bid and the lowest ask are the best quotes, their average is the *mid price* and their difference the *spread*; the size at the first levels on each side is the *depth*. Prices on Polymarket live in $[0, 1]$ with a tick of 0.001 on most markets, so a contract that is almost certain trades at 0.99 or above and its price can only move by a few ticks. Because the Yes and No books mirror each other, the book of one side describes the market, and this thesis reads the Yes side. An *order-book event* is a change of the size resting at one price on one side; most events do not touch the best quotes.

[TODO: the two consequences that recur in every result: a bounded, discrete price whose changes are mostly one-tick oscillations that revert, and dynamics that depend on the price level. The microstructure study of Dubach [8].]

2.2.4 The life of a market

[TODO: creation and metadata (known in advance), trading, the end date; the crypto interval markets that live five or fifteen minutes, sports before and during the game, political questions for months; resolution through UMA's optimistic oracle [9], which happens after the end date and is never an input]

2.2.5 What is known about Polymarket

[TODO: related work: outcome forecasting with language models and news; dataset papers; the economics of the platform. The closest work is Young [10] on the 15-minute Bitcoin markets. This thesis predicts the price while the market is open, not the outcome. Move the citations from the old related-work section.]

2.3 Time series, order books and sequence models

2.3.1 Concepts

[TODO: to write. Outline:]

- time series; the order-book and transaction streams are sequences of timed *events*, not series sampled on a regular grid;
- two ways of reading a stream: a fixed summary of the past handed to a learner without memory (*stateless*), or a *state* per market updated at every event (*stateful*, recurrent); what each can see, how each is trained, how each handles irregular time;

- the task: the direction of the change over a horizon H, three classes (DOWN, CONST, UP); the random walk as the null model.

2.3.2 Price predictability and limit order books

The efficient-market hypothesis in its weak form states that past prices carry no information about future prices [11]. At short horizons this is only approximately true: the stylised facts of high-frequency returns [12] include a negative autocorrelation at the shortest lags, produced by the bid–ask bounce and by the discreteness of prices, and in a limit order book the next move of the mid is related to the order-flow imbalance at the best quotes [13]. [TODO: deep learning on limit order books, from DeepLOB [14] on; predictability depends on the microstructure more than on the architecture [15]; one model pooled over many assets [16]; the warning about linear baselines [17]]

2.3.3 Recurrent networks and multi-rate memories

[TODO: GRU [18] and truncated backpropagation through time; the clockwork RNN [19] and GRU-D [20] as the references of the hierarchical memory and of the continuous-time decay]

2.3.4 Time-series foundation models

[TODO: Chronos and Chronos-2 [21, 22]; the evidence in finance, close to a random walk]

2.4 Graphs and graph neural networks

2.4.1 Concepts

[TODO: to write. Outline:]

- graphs; the bipartite graph between wallets and markets; degree, components, k-cores;
- dynamic graphs: a sequence of snapshots or a stream of timed events;
- message passing; on a bipartite dynamic graph, from wallets to the markets they trade and back.

2.4.2 Graph neural networks

[TODO: message passing [23–26]; neighbourhood sampling and attention over the neighbours; a GRU as the node update [27]]

2.4.3 Dynamic graphs

[TODO: continuous-time dynamic graphs [28, 29]; the memory-based models up to Temporal Graph Networks [30]; a model should contain the strong heuristic as a special case [31]]

2.5 Gradient boosting and neural tabular learners

[TODO: LightGBM [32] as the reference on tables of features; how a neural tabular model has to be trained to compete [33, 34]; TabM [35]; logit adjustment under class imbalance [36]]

2.6 Evaluation under imbalance and dependence

[TODO: balanced accuracy [37]; dependent samples and the hour-block bootstrap [38]; the focal loss [39]]

Chapter 3

The Dataset: Collection, Preparation and Exploration

This chapter follows the data from their three sources to the objects the models read: what was collected and from where (Sections 3.1 and 3.2), how large and how reliable it is (Sections 3.3 and 3.4), how the raw streams were preprocessed (Section 3.5) and how the final dataset was defined (Section 3.6). It closes with an exploratory and statistical analysis of the data (Section 3.7).

3.1 The four tables

The dataset covers twenty-one consecutive days, from 6 to 26 March 2026, and consists of four tables (Table 3.1). The *markets* table describes every market that existed in the period: question text, tags, the two outcome tokens, creation and end dates, and the final outcome. The *order book* table is the stream of price-level updates of the "Yes" side of every market's book: each row carries the best bid and ask, the changed price level, size and side, and the mid price. The *snapshots* table holds full order-book states. The *transactions* table is the on-chain record of every fill: block number, timestamp, transaction hash, wallet, role (maker or taker), token, side, quantity, price and fee.

3.2 Collection

3.2.1 Market metadata: the Gamma API

The *markets* table was built from Polymarket's public metadata service (the Gamma API): the markets closed in the window were paged through its keyset endpoint, and the markets that appear in the transactions but not in that listing were completed one by one through the markets-by-token endpoint. The tags are the source of the eight categories used throughout the thesis (crypto, sports, esports, politics, weather, economy and tech, culture, other).

3.2.2 The order-book archive

The *order book* and *snapshots* tables come from a public archive of the operator's websocket feed published on Kaggle [40]: one file per day, restricted to the Yes side of every book, with the creation time and the reception time of every message.

Table 3.1: Data sources. Row counts from the DuckDB database.

Table	Rows	Content
markets	175,476	one row per market: text, tags, tokens, dates, outcome
order book	5,555,777,555	price-level updates of the Yes-side book
snapshots	51,674,425	full order-book states (JSON)
transactions	218,882,492	on-chain fills with wallet, token, side, size, price

3.2.3 The on-chain transactions: a blockchain scraper

The *transactions* table was collected with a modified version of `poly-trade-scan` [41], an open-source client that reads every Polygon block from a public node, filters the transactions addressed to Polymarket's exchange contracts and decodes their `matchOrders` calls; the modified version emits one row per wallet and per fill. [TODO: details of the decoding (taker and maker legs, fees), the node, the retries; 219 million rows are fill legs]

3.3 Scale, coverage and storage

Daily activity is stable over the period: 9.2–11.8 million transaction rows, 122,000–161,000 distinct wallets and 15,000–21,000 active markets per day. The data are stored as Parquet files and queried with DuckDB [42]. [TODO: storage sizes; a figure of the daily volumes]

3.4 Data quality

[TODO: the facts about the data that shape every later choice, with the numbers:]

- fabricated mid on one-sided books (the feature builder substitutes 0.49 or 0.51 for the missing side);
- token orientation: the book refers to token 1 in Yes/No markets and to token 2 in all the others, while the raw trade direction refers to token 1 everywhere;
- about 40% of the samples have a mid below 0.05 or above 0.95;
- one dip in the transaction stream, on 20 March.

3.5 Preprocessing and audit of the pipeline

3.5.1 Time axis: blocks and the synchronisation of the two streams

Polygon produces a block roughly every two seconds, and every transaction carries its block number. The block is the time step of the models: a step is one block, the set of transactions in it, and the order-book updates that arrived since the previous block. Because the two streams are collected by different clocks, a model that acts at block t sees every book update created strictly before t and the fills of the block itself.

3.5.2 Derived tensors

[TODO: to write:]

- a 52-dimensional feature vector per market and per order-book update (best quotes, five levels per side, spread, mid, imbalances, recent trade statistics), stored in hourly partitions;

- the market lifecycle table: deployment time (known in advance) and last trade (hindsight only, used to define the samples and never as an input);
- static market features: a sentence embedding of the question and a few numeric fields;
- dense indices for markets and wallets.

3.5.3 Audit of the pipeline

[TODO: the three errors found by the audit and fixed before any result (wrong price column; leaky momentum baseline; incomplete label lookup), and what was retrained]

3.6 The final dataset: labels, samples, features and periods

3.6.1 Labels

The target of a market m at block time t and horizon H is the sign of the change of its mid price:

$$y_{m,t} = \operatorname{sign}\big(\operatorname{mid}_m(t+H) - \operatorname{mid}_m(t)\big) \in \{\text{DOWN}, \text{CONST}, \text{UP}\}, \tag{3.1}$$

where $\operatorname{mid}_m(s)$ is the mid price of the last order-book update of m strictly before s. Three classes, because over one hour the mid of more than half of the samples does not move at all; no dead zone, any move of at least one tick counts; when $t+H$ is beyond the loaded data the label is missing. [TODO: samples: one per (market, block) with an update; the deduplicated view, one sample per market per fifteen minutes]

3.6.2 Feature engineering

[TODO: to write. The feature tables:]

- the sixty-one hand-made causal features, in five groups: price history, book shape, activity and clock, on-chain trades, static fields of the market (the full list in Appendix A);
- sibling features: the contracts of the same event;
- news features: GDELT counts and tone for the entities named in the question;
- wallet features on the transaction edges;
- forecast features: the quantiles of a zero-shot Chronos-2 forecast.

3.6.3 Training, validation and test periods

The twenty-one days are split by date: models are trained on 6–12 March and every design choice is taken on the validation day, 13 March; the period 24–26 March is a locked test set, never evaluated and never inspected, reserved for the final numbers.

3.7 Exploratory and statistical analysis

[TODO: in closing, as agreed: what the data say before any model is trained, in three parts. Numbers on the training and validation days only.]

3.7.1 The price process

[TODO: outline: the event stream and how uneven it is; how often a price moves at each horizon (Table 3.2); the price level; the activity clock; the direction of a move; the kind of contract; the contracts of the same event; the wallets]

Table 3.2: Share of the three classes among the labelled validation samples, by horizon.

Horizon	DOWN	CONST	UP
1 min	0.057	0.891	0.052
15 min	0.164	0.690	0.146
1 h	0.236	0.549	0.215
6 h	0.335	0.354	0.312
12 h	0.377	0.275	0.348

3.7.2 The transactions as a graph

[TODO: size, density and degree distributions; components and k-cores; the graph in time; a map of the graph (Fig. 3.1); synchronous publics; what it says for the models]

3.7.3 The structure of the feature space

[TODO: data-mining study of the hand-made features: how many quantities they contain (PCA, correlation blocks); maps of the samples (UMAP, t-SNE); linear, local and threshold separability; families of markets; across the horizons]

3.7.4 What the data say before a model is trained

[TODO: summary that sets the agenda of the methodology]

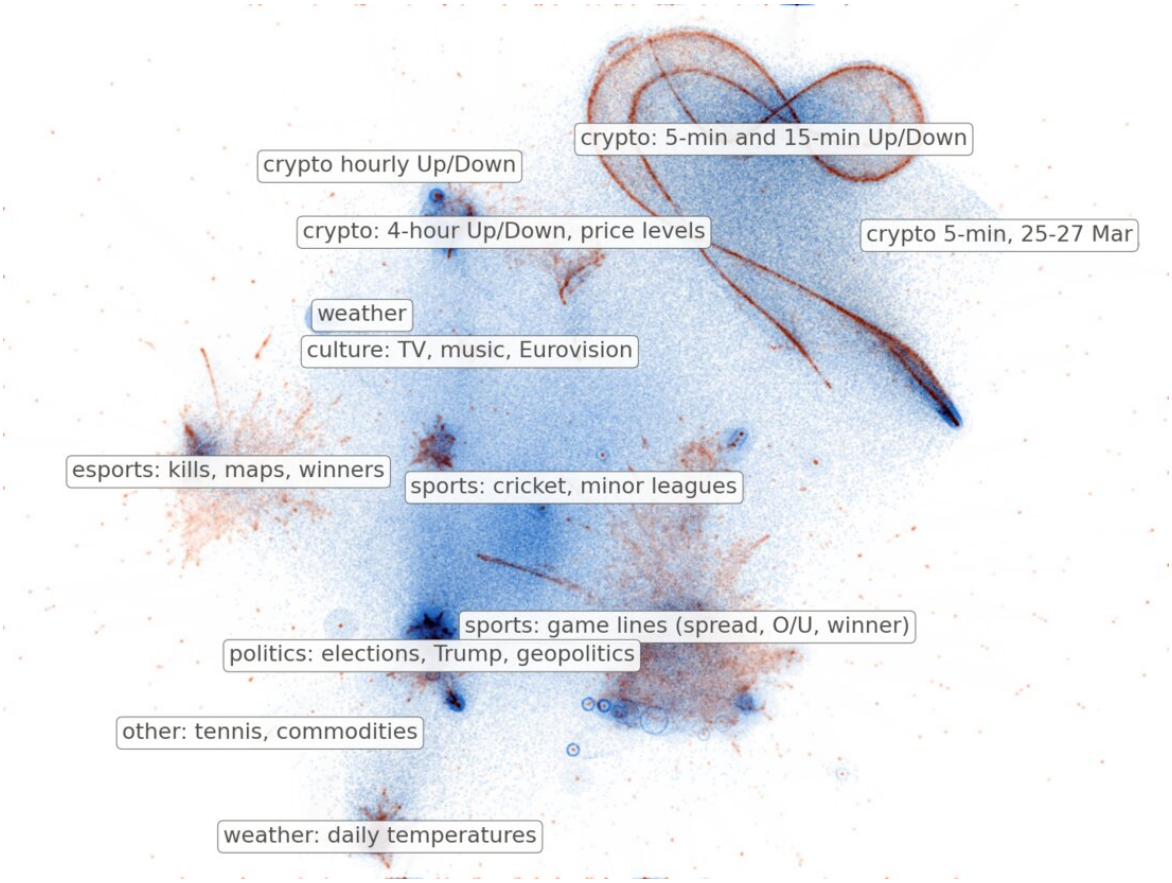


Figure 3.1: Three weeks of transactions as a graph: wallets (blue) and markets (orange), joined by the trading relations between them, 6–26 March 2026. [TODO: caption]

Chapter 4

Methodology

This chapter sets out the rationale of the study (Section 4.1) and the approaches that were compared, in five families: causal rules on the price, stateless learners on hand-made features, sequence models of the order-book stream, graph models, and sources of information outside the book. [TODO: each family closes with its advantages and limits a priori; a summary table at the end]

4.1 Objective, criteria and rules of inclusion

4.1.1 Objective

The object of the study is the direction of the mid price of a Polymarket market over a horizon H, read as the three-class label of Section 3.6.1, at every block in which the market updated its book, for five horizons from one minute to twelve hours. The task is the prediction of the price while the market is open, not of the outcome of the event; the measure is balanced accuracy, because the three classes are of very unequal size and change size with the horizon; and the unit of every comparison is the paired difference between two models on the same samples, with an interval.

4.1.2 Criteria of success

A model is said to read information that another does not when its balanced accuracy on the validation day exceeds the other's on the same samples, after the same post-hoc calibration, by a margin whose 95% hour-block bootstrap interval excludes zero. The reference competitor is first the best causal rule found on the data and then, once it exists, the best stateless learner on hand-made features. [TODO: ties; the two views of the samples (full stream and deduplicated); the locked test period]

4.1.3 Rules of inclusion and exclusion

[TODO: to expand, one paragraph each:]

1. **Causal inputs only.** Everything an approach reads at time t is computed from data strictly before t, and every label from data after it.
2. **Public information that the dataset contains**, plus two public external sources that could be aligned to it (the news flow, a pretrained forecaster); no external reference prices.
3. **One model for all markets**, never one model per market.
4. **Every approach is measured against the best simpler one on the same inputs**, and where possible contains it as a special case.

5. **Families that span the questions**: each research question has a paired comparison. Out of scope: outcome forecasting with language models, trading strategies, per-market models.
6. **A fixed compute budget**: one run per GPU, hours per run; short tests to screen the variants, full-length runs behind every claim.

4.2 Baselines: causal rules on the price

All baselines are causal rules on the mid price of the market itself. With $\Delta_L^- = \text{mid}(t) - \text{mid}(t-L)$ the change over the past lookback L:

- **Constant**: always CONST; balanced accuracy 1/3 by construction.
- **Momentum-L**: predict $\text{sign}(\Delta_L^-)$.
- **Mean Reversion-L**: predict $-\text{sign}(\Delta_L^-)$, and CONST if the price did not move.

Advantages and limits. [TODO: to write]

4.3 Stateless learners on hand-made features

A LightGBM classifier [32] on the sixty-one features of Section 3.6.2, trained with class-balanced weights on 7–11 March and early-stopped on 12 March, adding one group of features at a time. On the same table, an MLP and a TabM ensemble, trained as tabular models should be (Section 2.5), to tell whether a difference between the trees and a deep model is the learner or the training regime. [TODO: hyper-parameters]

Advantages and limits. [TODO: to write]

4.4 Sequence models of the order-book stream

4.4.1 The continuous-time stateful model

The model (CT-DGNN in the code) processes the stream block by block. At block time t it receives the markets that updated their book since the previous block, with a window of their last K book states, and the transactions of the block as edges between wallets and markets; it keeps a state vector for every market and every wallet. [TODO: one step of the model, with the architecture figure from the old chapter:]

- states and initialisation; continuous-time decay of the states;
- order-book update: the window encoders (space first, or time first);
- hierarchical memory at three rates; global state;
- long-lookback price features and the direct path, which contains the mean-reversion rule as a special case;
- predictor, loss and training (truncated BPTT, one pass through the stream).

4.4.2 The stateful model as the residual of the trees

[TODO: cross-fitted tree log-probabilities added to the output logits; at initialisation the model is the tree model, and the number that measures the state is the gain over it]

4.4.3 A pretrained forecaster used zero-shot

[TODO: Chronos-2 on the last 512 mids of the market; three decision rules; the same quantiles as features of the stateful model]

Advantages and limits. [TODO: to write]

4.5 Graph models

4.5.1 Message passing on transactions

[TODO: edges (w, m) with a 7-dimensional feature; attention over the incident edges and a GRU update; the token-orientation feature; the enrichments (wallet skill, placebo skill, past behaviour); the ablation without edges]

4.5.2 The two graphs

[TODO: the wallet–market graph, read by message passing, and the market–market graph between the contracts of the same event, measured first through the sibling features]

Advantages and limits. [TODO: to write]

4.6 Information outside the book

[TODO: the foundation-model forecasts, the news flow, the text of the market; each measured first in the stateless learner]

Advantages and limits. [TODO: to write]

4.7 The families at a glance

[TODO: table: family, members, what it reads, where its state lives, reference]

Chapter 5

Experiments and Results

[TODO: the results of the old chapters, reorganised by family in the order of Chapter 4; only the setup and the causal rules moved so far]

5.1 Experimental setup

Models are trained on 6–12 March and validated on 13 March, with 24–26 March locked for the final numbers. All numbers are balanced accuracies on the validation day, reported on all samples and on three subsets: the *deduplicated* samples (at most one per market per fifteen minutes), the *extreme* ones (mid outside $[0.05, 0.95]$) and the *central* ones. Differences between models are computed on exactly the same samples and carry a 95% hour-block bootstrap interval. [TODO: the three rules that make the comparisons fair: the same samples; the same post-hoc calibration (two logit offsets, cross-fitted on the validation hours); full-length runs behind every claim. Compute: one run per GPU, hours per run.]

5.2 Rules and stateless learners

5.2.1 Causal rules

Mean Reversion-H is above 0.5 at every horizon and it is the rule to beat (Table 5.1); Momentum is far below it. [TODO: why mean reversion is strong: a bounded, discrete price whose moves are mostly one-tick oscillations that revert]

5.2.2 Trees on hand-made features: the no-state control

[TODO: the table by feature group at one hour; the other horizons; where the information is]

5.2.3 Neural tabular learners: trees against networks

[TODO: MLP and TabM against LightGBM, three seeds]

5.3 Stateful models

[TODO: from the first models to the last: the retraining on the correct target; loss and imbalance variants; memory, direct path and lookbacks; the window encoders; the state given the control's inputs; the cost of the online regime; the residual of the trees; what the state is worth]

Table 5.1: Causal rules on the validation day, balanced accuracy, all samples and deduplicated samples. Momentum-H and Mean Reversion-H use the change over the past H; Momentum last-tick uses the sign of the last mid change.

	1 min		15 min		1 h		6 h		12 h	
Rule	all	dedup	all	dedup	all	dedup	all	dedup	all	dedup
Constant	0.333	0.333	0.333	0.333	0.333	0.333	0.333	0.333	0.333	0.333
Momentum last-tick	—	—	—	—	0.375	0.387	0.365	0.374	0.362	0.370
Momentum-H	—	—	—	—	0.472	0.445	0.436	0.422	0.424	0.420
Mean Reversion-H	**0.536**	**0.501**	**0.537**	**0.525**	**0.527**	**0.506**	**0.501**	**0.488**	**0.507**	**0.500**

5.4 Graph models

[TODO: the wallet–market graph at every stage of the design; smart money; the sibling contracts; why the graph does not help]

5.5 Information outside the book

[TODO: foundation models zero-shot and as features; news volume; the text of the market]

5.6 Across horizons, kinds of market and seeds

[TODO: one and fifteen minutes; one hour; six and twelve hours; the synthesis table; by kind of market; robustness (seeds, calibration, short tests against full runs)]

5.7 Test set

[TODO: final numbers on 24–26 March, produced once at the end]

5.8 Discussion

[TODO: the results against the four questions of Section 1.3]

Chapter 6

Conclusions

[TODO: after the results chapter]

6.1 What the thesis found

[TODO: question by question]

6.2 Limitations

[TODO: to write]

6.3 Future work

[TODO: to write]

Appendix A

The hand-made features

[TODO: table of the sixty-one features, one line each, with the definition and the group (price history, book shape, activity and clock, trades, static)]

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